CURRICULUM VITAE

Darah N. Oxford

PhD Candidate · University of Missouri oxforddn@missouri.edu Columbia, MO

ABOUT

Profile

A versatile scientist and educator whose background spans animal communication, neuroethology, physiology, ecology, entomology, and nursing science. Brings PhD-level research in neuroethology, hands-on clinical nursing experience across ICU, cardiac, and rehabilitation settings, and a demonstrated record of innovative course design. Deeply invested in active learning, undergraduate research mentorship, and the practical integration of AI tools into evidence-based pedagogy.

Intro & General Biology · Human Physiology & Anatomy · Neuroscience & Animal Behavior · Ecology & Entomology · Research Methods & Biostatistics · Nursing & Health Sciences · AI Integration in Higher Ed · Science of Learning · Plant Biology & Pollinator Ecology

CREDENTIALS

Education

PhD, Biological Sciences
University of Missouri – Columbia

Expected May 2027 · Adviser: Dr. Johannes Schul
Dissertation: Neural circuits underlying call pattern generation in *Neoconocephalus* katydids
Pillars of Discovery Writing Fellow, 2026

BS, Nursing
Southern Arkansas University

May 2018 · Summa Cum Laude · GPA 3.92
Sigma Theta Tau · President's List: Fall 2016, Spring 2017, Fall 2017

MS, Entomology
University of Missouri

BS, Biology
University of Missouri

May 2005 · Cum Laude · GPA 3.75

December 2010 · GPA 3.6
Native plant characteristics & insect load

EXPERIENCE

Professional Activity

Guest Lecturer — Integrative Psychology (PSYC 495)

Spring 2024 & 2026

Columbia College, MO

- Statistical Design Methods for senior capstone; invited to return based on student and faculty feedback.

Course Developer & Lab Instructor — Human Physiology (Bio 3700)

2023–2026

University of Missouri

- Designed the new laboratory course (Summer 2023): lesson plans, instructor resources, student assessments, and Canvas LMS site; first offered Fall 2023.
- Taught two lab sections per semester in Fall 2023, Spring 2024, and Fall 2026 (Fall 2026: 24 students each, mostly upper-level pre-health); integrates clinical nursing experience (ICU, cardiac cath/EP lab, rehabilitation) into physiology instruction.

Graduate Teaching Assistant — Bio 1500

Aug 2020–May 2023

University of Missouri

- Lab & discussion sections: team-based learning, health equity, scientific reasoning, data interpretation.
- Teaching effectiveness scores 4.72–4.80 / 5.0; 76th–89th percentile among all Division of Biological Sciences graduate TAs.

Camp Co-Director & Science Lead — STEAM Summer Camp

July 2023

Columbia College, MO

- Week-long camp for grades 3–8; led the science program: flower and insect anatomy under dissecting microscopes, pollination, color and light (chromatography), and how bees see ultraviolet light.
- Built a partnership of 15 local organizations and businesses, including sponsors 3M and Veterans United and presenters from the Missouri Department of Conservation, the Columbia Aeronautics and Space Association (CASA) student aerospace and robotics club, Columbia College nursing simulation, and a local beekeeper.

Mini-Course Instructor — Columbia STEM Alliance Summer Expedition

July 2021

Columbia, MO

- "Pretty Plants & Yucky Bugs?" — co-taught with Dr. Jon Oxford: angiosperms and gymnosperms, plant-pollinator interactions, and entomology; hands-on flower dissections and microscopy.

Biology Lab Teacher — BIOME Intensive Orientation Experience

August 2022

University of Missouri

- Lab instruction for incoming first-year students.

Peer Tutor — Nursing Program

2017–2018

Southern Arkansas University

- Individualized and group tutoring; nursing skills labs; pharmacology calculations.

Supplemental Instructor — World Literature II

2016–2018

Southern Arkansas University

- Weekly discussion and exam prep sessions. Supplemental Instructor of the Year (2017); SI of the Month (Oct 2016, Feb 2017).

Substitute Teacher

2011

Columbia Public Schools

- Extended full-time assignments; elementary, secondary, special education.

ESL / English Conversation Instructor

2005–2007; 2012–2014

International Church, MU Campus & Community Center, Germany

- English conversation classes for international students and German community members.

RESEARCH

Doctoral Research — Neuroethology

2020–Present

University of Missouri

- Identifying neural circuits underlying call generation in *Neoconocephalus* katydids via neuroanatomy, electrophysiology, and behavioral ecology.
- Supervised two undergraduate research contributions: CNS circuit identification poster (J. Kelly-Cuenca, 2023) and wing coordination poster (B. Clemons, 2023).

Master's Research — Entomology & Plant Sciences

2005–2010

University of Missouri

- Native plant characteristics and insect load; plant-pollinator interactions; experimental field and lab design.

Hormone Assay Laboratory

June–Sept 2010

University of Missouri

- Enzyme-linked immunoassay: salivary testosterone and cortisol.

CLINICAL — REGISTERED NURSE (MISSOURI LICENSED)

Registered Nurse — Rusk Rehabilitation Hospital

Mar–Jul 2020

Columbia, MO

- Stroke, TBI, hip fracture, spinal cord, and amputee rehabilitation. Individualized care plans; interdisciplinary team; patient and caregiver education.

Registered Nurse — Lutheran Senior Services REACH Rehabilitation

Jan–Aug 2020

Columbia, MO

- Person-centered care; post-acute rehabilitation unit.

Registered Nurse — Cardiac Catheterization / Electrophysiology Lab

Aug–Dec 2019

University Hospital, Columbia, MO

- EP procedures: ablations, cardioversion, pacemaker/ICD implants, electrophysiology studies.

Registered Nurse — ICU

Jun 2018–Jun 2019

Magnolia Regional Medical Center, Magnolia, AR

- Total ICU care, telemetry, outpatient procedures; floated to Emergency Room, Med-Surg, Surgery.

AI & SCHOLARSHIP OF TEACHING

Workshop Leader — Virtual Conference on Teaching (VCT)

Feb 2026

Society for the Teaching of Psychology (APA Div. 2)

- "Unlocking AI's Potential in Higher Ed" — 2-hour interactive workshop with live demos: agentic browser detection, split-screen assessment systems, discipline-specific course chatbots, interactive learning modules; drew on experiments documented in the Imagination is the Key database.

Co-Creator & Curator — Imagination is the Key ↗

2025–Present

Public AI-in-Teaching Database (with Dr. Jon Oxford, Columbia College)

- Co-created and maintain a public, searchable database of 200+ real AI-in-teaching experiments, organized by topic, difficulty, and return on investment; anchors faculty AI workshops, open office hours, and consultations.

SCHOLARSHIP

Publications & Presentations

Oxford, D. N., & Oxford, J. (In Preparation). Imagination Is the Key: AI experiments as faculty development. *Society for the Teaching of Psychology, Teaching Tips*. Invited following ACT 2026.

Oxford, J., & Oxford, D. N. (In Preparation). Blueprint First: Building an AI-assisted assessment ecosystem. *Society for the Teaching of Psychology, Teaching Tips*. Invited following ACT 2026.

Oxford, J., & Oxford, D. N. (2026). Imagination is the Key: A public database of daily AI experiments in teaching. *Society for the Teaching of Psychology, Teaching Tips*.

Oxford, D. N., & Oxford, J. (2026, October 16). Imagination Is the Key: AI experiments as faculty development [Poster]. Annual Conference on Teaching (ACT), Society for the Teaching of Psychology, St. Louis, MO. *Interactive poster on learning from public, documented AI experiments, from custom chatbots to AI agents, following one experiment (an AI browser taking a course quiz) through to a redesigned assessment.*

Oxford, J., & Oxford, D. N. (2026, October 16). Blueprint First: Building an AI-assisted assessment ecosystem [Symposium]. Annual Conference on Teaching (ACT), Society for the Teaching of Psychology, St. Louis, MO.

Oxford, J., & Oxford, D. N. (2026). Imagination is the Key: A public database model for faculty AI development [Poster, accepted; not presented]. AAC&U Conference on AI and Higher Education.

Oxford, J., & Oxford, D. N. (2026). Hamilton-Zuk Hypothesis. In Shackelford, T. K. (Ed.), *Encyclopedia of Sexual Psychology and Behavior*. Cham: Springer International Publishing.

Oxford, J., & Oxford, D. N. (2022). Testosterone: Digit ratios. In Shackelford, T. K. (Ed.), *Encyclopedia of Sexual Psychology and Behavior*. Springer.

Oxford, J., & Oxford, D. N. (2022). Testosterone: Paternal investment. In Shackelford, T. K. (Ed.), *Encyclopedia of Sexual Psychology and Behavior*. Springer.

Oxford, J., & **Oxford, D. N.** (2017). Co-sleeping. In Shackelford, T. K., & Weekes-Shackelford, V. A. (Eds.), *Encyclopedia of Evolutionary Psychological Science*. Cham: Springer. *(Coauthored as a B.S.N. student.)*

Schul, J., Murphey, K., **Oxford, D.**, Kelly-Cuenca, J., & Favela, E. (2021). How does the central nervous system change when new call types evolve? Talk, Virtual Meeting of the Animal Behavior Society.

Oxford, D. N. (2018, April). Holding via kangaroo care provides multiple benefits to preterm low birth weight infants: Cardiorespiratory stability, weight gain, less incidence of hypothermia and decreased pain [Poster & podium presentation]. UAMS Nursing Research Conference. *Winner: Best BSN Poster & Podium Presentation.*

RECOGNITION

Awards & Honors

- Featured in *The Buzz*, Department of Biological Sciences digital magazine — Schul lab research feature highlighting dissertation work (July 2026)
- Pillars of Discovery Summer Dissertation Writing Fellowship — Graduate School, MU (2026)
- Sandra K. Abell Science Education Award — Outstanding graduate student with dual passion for research and college science teaching (April 2024)
- Griffiths Leadership Society for Women — Inducted Fall 2020
- Dean's List, UAMS College of Nursing Nurse Practitioner Program (Spring 2019)
- Best Student Research Poster & Podium Presentation, UAMS Nursing Research Conference (2018)
- Supplemental Instructor of the Year Award & Scholarship (2017)
- Sigma Theta Tau, Nursing Honor Society — Inducted October 2017
- Southern Arkansas University President's List (4.0 GPA): Fall 2016, Spring 2017, Fall 2017

SERVICE & MEMBERSHIP

Professional Memberships & Service

- Open Office Hours with the Oxfords: co-facilitator, drop-in AI support for faculty (Spring 2026–Present)
- Graduate Coalition Alliance (Spring 2022–Present)
- Sigma Theta Tau, Nursing Honor Society (2017–Present)

- Biological Graduate Student Association (2020–Present)
- Coalition of Graduate Workers (Fall 2020–Present)
- Griffiths Leadership Society for Women (2010, 2020–Present)
- Vice President of Nursing (Class of 2018), SAU (2016–2018)
- Arkansas Student Nurses Association, Projects Manager (2017–2018)
- C.V. Riley Entomological Society, Public Relations Officer (2009)

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